# PRACTICAL NO.1(A)

**Aim:-** Introduction to Business Aanalysis(EDA and customer Data Analysis).

## Code:-

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df=pd.read_csv("Practical_1.csv")
df.head()
```

Press **Shift** + **Enter** to execute the code.

## Output:-

| | Invoice_ID | Branch | City | Customer Type | Gender | Product Line | Unit Price | Quantity | Total | Date |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | INV001 | A | Mumbai | Member | Female | Food & Beverages | 650 | 2 | 1300 | 02-02-2026 |
| 1 | INV002 | B | Mumbai | Normal | Male | Electronics | 486 | 4 | 1944 | 26-03-2026 |
| 2 | INV003 | C | Pune | Member | Female | Fashion | 1475 | 10 | 14750 | 02-04-2026 |
| 3 | INV004 | A | Pune | Normal | Male | Health & Beauty | 574 | 5 | 2870 | 19-05-2026 |
| 4 | INV005 | B | Mumbai | Member | Female | Food & Beverages | 567 | 3 | 1701 | 02-06-2026 |

df.info()
(Gives all the information like null value count and data type of each column).

## Output:-

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 10 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Invoice_ID     5 non-null      object
 1   Branch         5 non-null      object
 2   City           5 non-null      object
 3   Customer Type  5 non-null      object
 4   Gender         5 non-null      object
 5   Product Line   5 non-null      object
 6   Unit Price     5 non-null      int64
 7   Quantity       5 non-null      int64
 8   Total          5 non-null      int64
 9   Date           5 non-null      object
dtypes: int64(3), object(7)
memory usage: 532.0+ bytes
```

df.isnull().sum()
(Give the total number of null values present in each column).

**Output:-**

```
Invoice_ID       0
Branch           0
City             0
Customer Type    0
Gender           0
Product Line     0
Unit Price       0
Quantity         0
Total            0
Date             0
dtype: int64
```

df.describe()
(Gives all the descriptive statistics like mean, std dev, min, max, 25%, 50%, 75%).

**Output:-**

| | Unit Price | Quantity | Total |
|---|---|---|---|
| **count** | 5.000000 | 5.000000 | 5.000000 |
| **mean** | 750.400000 | 4.800000 | 4513.000000 |
| **std** | 409.202028 | 3.114482 | 5751.653936 |
| **min** | 486.000000 | 2.000000 | 1300.000000 |
| **25%** | 567.000000 | 3.000000 | 1701.000000 |
| **50%** | 574.000000 | 4.000000 | 1944.000000 |
| **75%** | 650.000000 | 5.000000 | 2870.000000 |
| **max** | 1475.000000 | 10.000000 | 14750.000000 |

df["Customer Type"].value_counts()
(Counts the total number customer type in dataset).

**Output:-**

```
Customer Type
Member    3
Normal    2
Name: count, dtype: int64
```

df["Gender"].value_counts()
(Counts the total number of customers in each type of gender).

**Output:-**

```
Gender
Female    3
Male      2
Name: count, dtype: int64
```

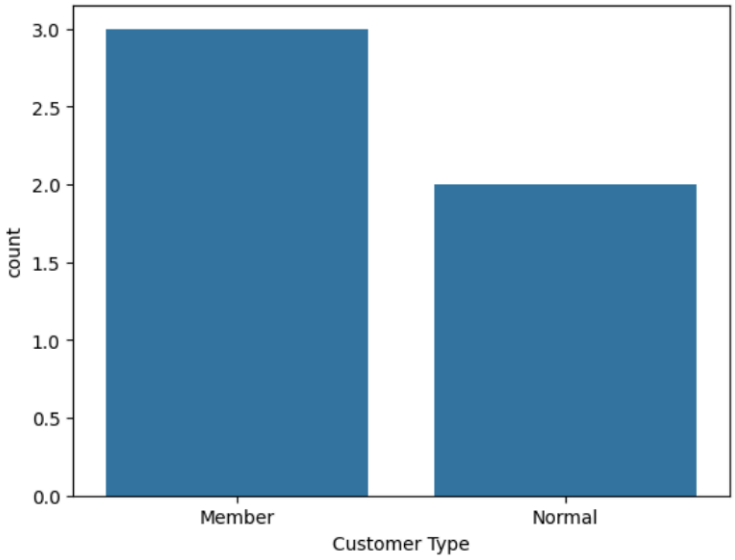
df["Product Line"].value_counts()
(It counts total number of product in each product line).

**Output:-**

```
Product Line
Food & Beverages    2
Electronics         1
Fashion             1
Health & Beauty     1
Name: count, dtype: int64
```

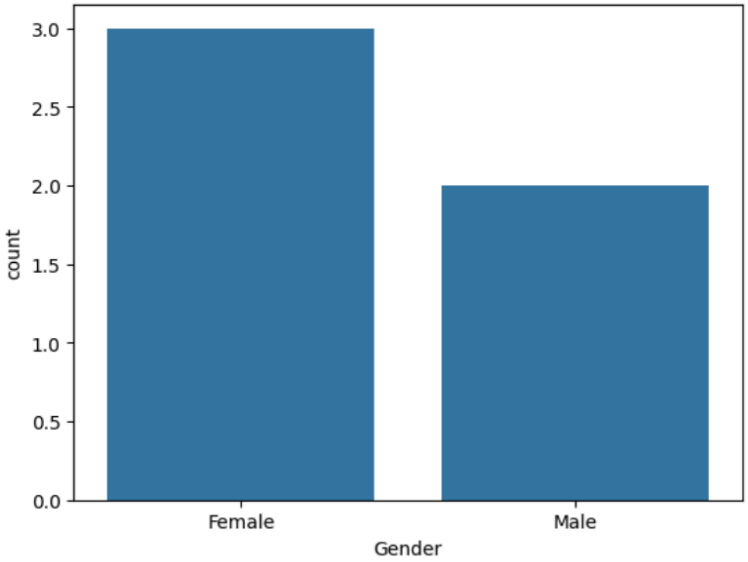
```
sns.countplot(x="Customer Type", data=df)
plt.show()
```

**Output:-**



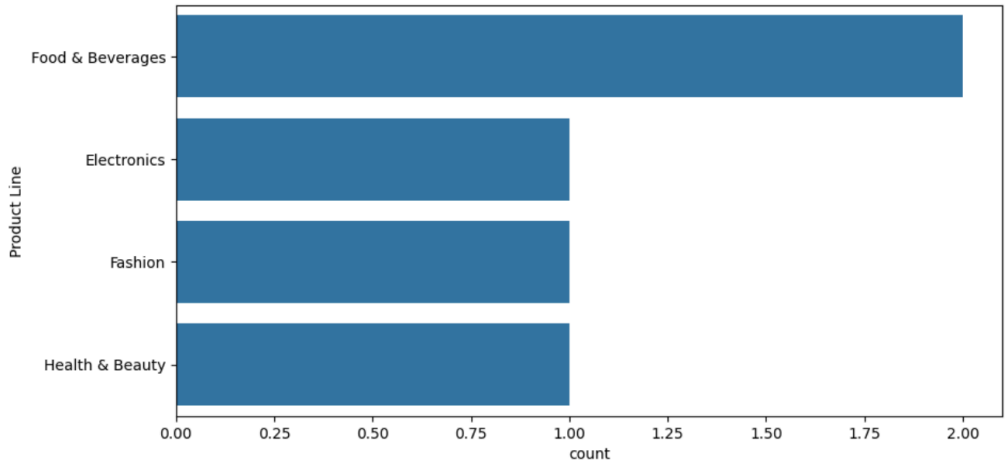

```
sns.countplot(x="Gender",data=df)
plt.show()
```

**Output:-**



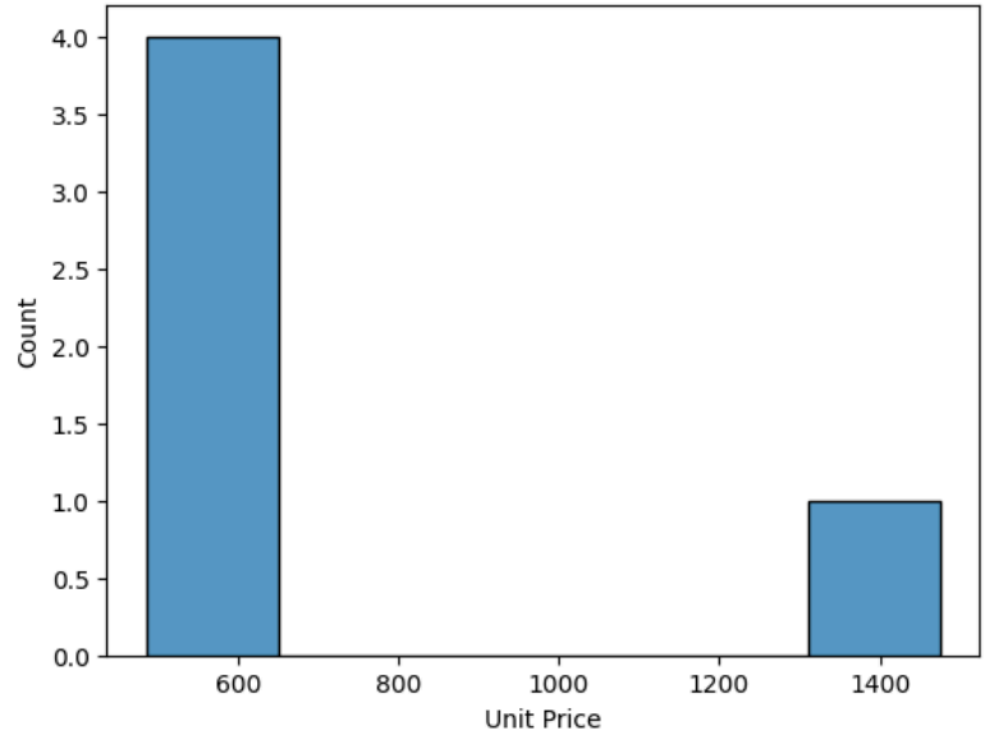
```
plt.figure(figsize=(10,5))
sns.countplot(y="Product Line",data=df)
plt.show()
```

**Output:-**



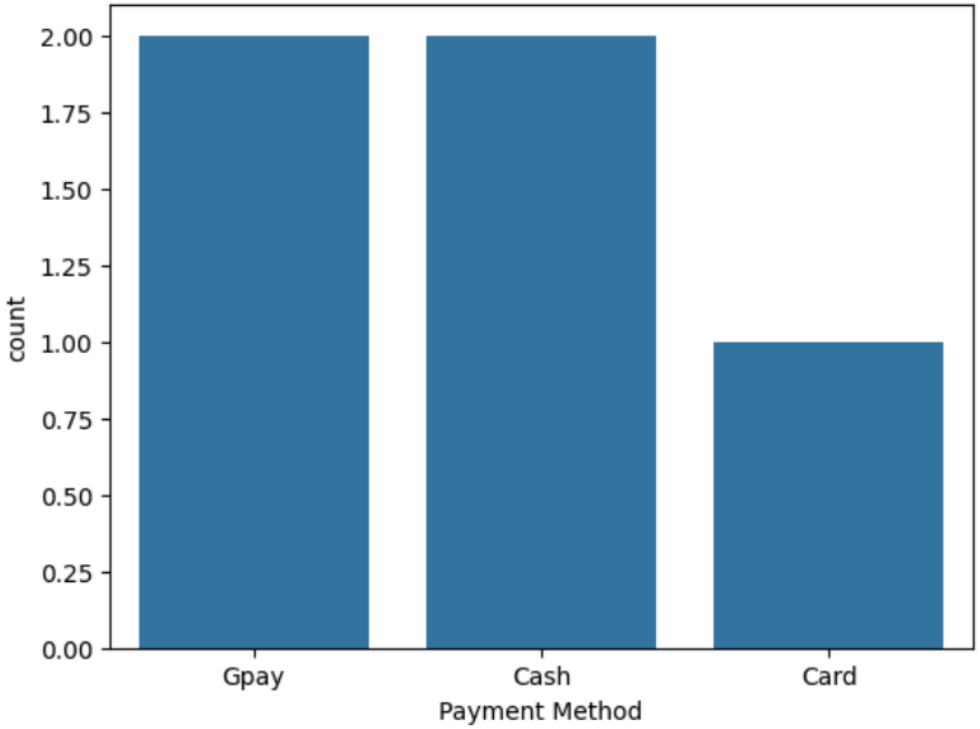

```
sns.histplot(df["Unit Price"],bins=6)
plt.show()
```

**Output:-**



```
sns.countplot(x="Payment Method",data=df)
plt.show()
```

## Output:-




## Conclusion:-

This practical introduced the basics of **Business Analysis using Data Analysis (EDA)**. The sales dataset was successfully loaded and analyzed using Python libraries such as Pandas, Matplotlib, and Seaborn. The dataset was checked for its structure, data types, and missing values, and no missing values were found. Statistical measures such as mean, minimum, maximum, and standard deviation were obtained using descriptive analysis. Customer type, gender, product line, unit price, and payment-related data were analyzed using different visualizations. Count plots and histograms helped in understanding customer and product distributions. Overall, this practical demonstrated how data analysis and visualization can provide useful insights for better understanding business and customer data.